

Resource-Constrained Project Scheduling Problem with Alternative Subgraphs (RCPSP-AS)

1 Problem Definition

A project is represented as a directed acyclic graph $G = (N, A)$ where $N = \{0, \dots, n\}$ is a set of non-preemptive, topologically ordered activities and $A \subset N \times N$ is a set of end-start precedence arcs. Each activity i has a fixed duration d_i and demands $r_{i,v}$ units of each renewable resource $v \in R$, which has limited capacity a_v . Activity 0 is the unique source and activity n the unique sink.

RCPSP-AS extends RCPSP with a set of *alternative subgraphs* L . Each subgraph $l \in L$ has a principal activity p_l (source of the subgraph), a terminal activity t_l (sink), and a set of branches K_{p_l} . Each branch $k \in K_{p_l}$ starts from a *branching activity* b_k , an immediate successor of p_l , and contains activities $N_{b_k} \subset N$ with $b_k \in N_{b_k}$, $p_l \notin N_{b_k}$, and $t_l \notin N_{b_k}$. Activities belonging to at least one branch form the *alternative* set $N^a = \bigcup_k N_{b_k}$; the rest are *fixed*: $N^f = N \setminus N^a$. Exactly one branch must be selected from each subgraph whose principal activity is scheduled.

Two extensions are supported: *nested subgraphs* (a subgraph placed inside another's branch) and *linked branches*, indicated by $\kappa_{i,j} = 1$ when activity $i \in N_{b_k}$ links to $j \in N_{b_{k'}}$ ($k \neq k'$), causing the selection of i to imply the selection of j and its successors in branch k' .

The objective is to minimize the makespan $C_{\max} = s_n + d_n$.

2 Notation

3 Original Formulation

This formulation, following Servranckx & Vanhoucke (2019), distinguishes fixed and alternative activities explicitly and uses dedicated constraints for branch successor implications and inter-branch links.

3.1 ILP

$$\min \quad s_n + d_n \tag{1}$$

$$\text{s.t.} \quad x_i = 1 \quad \forall i \in N^f \tag{2}$$

$$\sum_{k \in K_{p_l}} x_{b_k} = x_{p_l} \quad \forall l \in L \tag{3}$$

$$x_{p_l} \leq x_{t_l} \quad \forall l \in L \tag{4}$$

$$x_i \leq x_j \quad \forall l \in L, k \in K_{p_l}, (i, j) \in (N_{b_k} \times N_{b_k}) \cap A \tag{5}$$

$$x_i \leq x_j \quad \forall (i, j) : \kappa_{i,j} = 1 \tag{6}$$

$$s_i + d_i \leq s_j + M(2 - x_i - x_j) \quad \forall (i, j) \in A \tag{7}$$

$$\sum_{\substack{i \in N: \\ s_i \leq t < s_i + d_i}} r_{i,v} \cdot x_i \leq a_v \quad \forall v \in R, t \in \mathcal{T} \tag{8}$$

$$x_i \in \{0, 1\}, \quad s_i \in \mathbb{Z}_{\geq 0} \quad \forall i \in N \tag{9}$$

Symbol	Description
<i>Sets and indices</i>	
$N = \{0, \dots, n\}$	Activities (topologically ordered); 0 = source, n = sink
$A \subset N \times N$	Precedence arcs
$A_{\text{prop}} = A \setminus \{(pl, j) : l \in L\}$	Non-branching arcs (simplified formulation)
L	Alternative subgraphs
K_{pl}	Branches of subgraph l
$N_{b_k} \subset N$	Activities in branch k
$N^a = \bigcup_k N_{b_k}$	Alternative activities
$N^f = N \setminus N^a$	Fixed activities
R	Renewable resources
$\mathcal{T} = \{0, \dots, T_{\max}\}$	Planning horizon (ILP)
$K = \{k^*\} \cup \bigcup_l K_{pl}$	All branches including dummy k^* (original CP)
$B_i = \{k \in K \mid i \in N_{b_k}\}$	Branches containing activity i (original CP)
<i>Parameters</i>	
d_i	Duration of activity i
$r_{i,v}$	Demand of activity i for resource v
a_v	Capacity of resource v
pl	Principal activity of subgraph l
tl	Terminal activity of subgraph l
b_k	Branching activity of branch k
$\kappa_{i,j} \in \{0, 1\}$	Link indicator: 1 if i in branch k links to j in branch $k' \neq k$
M	Sufficiently large constant (ILP)
<i>Decision variables — ILP</i>	
$x_i \in \{0, 1\}$	1 if activity i is selected
$s_i \in \mathbb{Z}_{\geq 0}$	Start time of activity i
<i>Decision variables — CP</i>	
x_i	Optional interval variable for activity i , size d_i
presenceOf(x_i)	1 if interval x_i is present, 0 otherwise
startOf(x_i)	Start time of interval x_i
endOf(x_i)	End time of interval x_i
pulse(x_i, h)	Contributes h to a cumulative function while x_i is present

(2) All fixed activities are always selected. (3) Exactly one branch is chosen per active subgraph. (4) If the principal activity is selected, so is the terminal. (5) Within a branch, selection propagates along precedence arcs. (6) Linked activities propagate selection across branches. (7) Big- M linearization of precedence: if both i, j are selected, i finishes before j starts. (8) Resource capacity is respected at every time point.

3.2 CP

Before building the model, a preprocessing step eliminates linked branches and unifies activity selection:

1. **Eliminate links.** For each $\kappa_{i,j} = 1$ between branches k and k' , extend $N_{b_k} \leftarrow N_{b_k} \cup (S_j^* \cap N_{b_{k'}})$, where S_j^* denotes the transitive successors of j .
2. **Dummy branch.** Introduce k^* with $b_{k^*} = 0$ and $N_{b_{k^*}} = N^f$.
3. **Derived sets.** $K = \{k^*\} \cup \bigcup_{l \in L} K_{pl}$ and $B_i = \{k \in K \mid i \in N_{b_k}\}$.

$$\min \quad \text{endOf}(\mathbf{x}_n) \quad (10)$$

$$\text{s.t.} \quad \text{presenceOf}(\mathbf{x}_0) = 1 \quad (11)$$

$$\text{presenceOf}(\mathbf{x}_i) \Leftrightarrow \bigvee_{k \in B_i} \text{presenceOf}(\mathbf{x}_{b_k}) \quad \forall i \in N \setminus \{0\} \quad (12)$$

$$\sum_{k \in K_{p_l}} \text{presenceOf}(\mathbf{x}_{b_k}) = \text{presenceOf}(\mathbf{x}_{p_l}) \quad \forall l \in L \quad (13)$$

$$(\text{presenceOf}(\mathbf{x}_i) \wedge \text{presenceOf}(\mathbf{x}_j)) \Rightarrow \text{endOf}(\mathbf{x}_i) \leq \text{startOf}(\mathbf{x}_j) \quad \forall (i, j) \in A \quad (14)$$

$$\sum_{i \in N} \text{pulse}(\mathbf{x}_i, r_{i,v}) \leq a_v \quad \forall v \in R \quad (15)$$

$$\mathbf{x}_i \text{ optional interval, size } d_i \quad \forall i \in N \quad (16)$$

After preprocessing, activity selection (12) is unified: activity i is present if and only if at least one branch containing it is selected. This single rule covers both fixed activities (via the dummy branch k^* tied to the always-present source) and alternative activities. Constraints (13)–(15) mirror the ILP: branch selection, conditional precedence timing, and cumulative resource limits.

4 Simplified Formulation

The simplified formulation replaces the explicit classification into N^f , N^a , t_l , and κ with a single mechanism: *selection propagation* along the set of non-branching arcs $A_{\text{prop}} = A \setminus \{(p_l, j) \in A : l \in L\}$. Since the source is always selected and A_{prop} contains all arcs except those from principal activities to their branch successors, fixed activities are selected by transitivity. Within a selected branch, successor selection follows the same propagation. Cross-branch links are handled automatically because the link arc (i, j) belongs to A_{prop} whenever the source i is not a principal activity. No preprocessing is needed.

4.1 ILP

$$\min \quad s_n + d_n \quad (17)$$

$$\text{s.t.} \quad x_0 = 1 \quad (18)$$

$$x_i \leq x_j \quad \forall (i, j) \in A_{\text{prop}} \quad (19)$$

$$\sum_{j: (p_l, j) \in A} x_j = x_{p_l} \quad \forall l \in L \quad (20)$$

$$s_i + d_i \leq s_j + M(2 - x_i - x_j) \quad \forall (i, j) \in A \quad (21)$$

$$\sum_{\substack{i \in N: \\ s_i \leq t < s_i + d_i}} r_{i,v} \cdot x_i \leq a_v \quad \forall v \in R, t \in \mathcal{T} \quad (22)$$

$$x_i \in \{0, 1\}, \quad s_i \in \mathbb{Z}_{\geq 0} \quad \forall i \in N \quad (23)$$

(18) The source activity is selected. (19) Selection propagates along non-branching arcs—this single rule replaces (2), (4), (5), and (6). (20) Exactly one successor of p_l is chosen iff p_l is selected. (21)–(22) Precedence and resource constraints are unchanged.

4.2 CP

$$\min \quad \text{endOf}(x_n) \quad (24)$$

$$\text{s.t.} \quad \text{presenceOf}(x_0) = 1 \quad (25)$$

$$\text{presenceOf}(x_i) \Rightarrow \text{presenceOf}(x_j) \quad \forall (i, j) \in A_{\text{prop}} \quad (26)$$

$$\sum_{j: (p_l, j) \in A} \text{presenceOf}(x_j) = \text{presenceOf}(x_{p_l}) \quad \forall l \in L \quad (27)$$

$$(\text{presenceOf}(x_i) \wedge \text{presenceOf}(x_j)) \Rightarrow \text{endOf}(x_i) \leq \text{startOf}(x_j) \quad \forall (i, j) \in A \quad (28)$$

$$\sum_{i \in N} \text{pulse}(x_i, r_{i,v}) \leq a_v \quad \forall v \in R \quad (29)$$

$$x_i \text{ optional interval, size } d_i \quad \forall i \in N \quad (30)$$

The CP formulation mirrors the simplified ILP directly. Optional interval variables (30) unify the selection (presenceOf) and timing (startOf, endOf) decisions. The pulse function (29) models time-varying resource usage, enabling the solver's built-in cumulative constraint.

5 References

Servranckx, T., & Vanhoucke, M. (2019). A tabu search procedure for the resource-constrained project scheduling problem with alternative subgraphs. *European Journal of Operational Research*, 273(3), 841–860.